Where Did That Idea Come From? Provenance Trails in Researchers' Note-Taking

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Abstract

Researchers write notes across many tools and later struggle to recall where an idea came from. In a three-week diary study with 19 researchers, we asked participants to trace the origin of a sample of their own notes. After two weeks, 31% of notes had no recoverable source; most of these originated in conversations, chat with AI assistants, or reading that was not saved. Participants valued provenance most when writing related work and when revisiting abandoned projects.

1 Introduction

A note records an idea; it rarely records the source. Researchers rebuild provenance when they need it, often at the worst time, such as when citing.

2 Method

19 researchers (PhD students to faculty, HCI, education and biology) kept a diary for three weeks. Twice a week, we sampled five of their notes from the previous days and asked them to find where each idea came from and how long it took.

3 Findings

Origin of note	Share of notes	Recoverable after 2 weeks
Paper or article read	34%	81%
Conversation (meeting, hallway)	22%	39%
Chat with AI assistant	17%	46%
Own earlier note	15%	88%
Talk or video	7%	52%
Unknown at time of writing	5%	-

Overall, 31% of sampled notes had no recoverable source after two weeks. Median search time for recoverable sources was 3.5 minutes.

AI chats were a growing and fragile source: participants rarely saved them, and those who did could not search across providers.

4 Implications

Tools should attach source context automatically at capture time, link notes to conversations and chats, and make provenance visible in the places researchers write (drafts, related-work sections). A single import of a researcher's existing tools is a precondition.

5 Limitations

Self-selected participants who already cared about note-taking; three weeks is short relative to a research project.

References

[1] Marchetti, E. Trails revisited. HT 2024, 201-210.

[2] Engelbart, D. C. Augmenting Human Intellect: A Conceptual Framework. SRI Summary Report AFOSR-3223, Stanford Research Institute, 1962.